«inference»

- hypo
  [Known]

$H_0: \mu = \text{value}$
$H_1: \mu > \text{(right)}$
$H_1: \mu \neq \text{(two)}$
$H_1: \mu = 0 \text{ (left)}$

- Test $= \dfrac{\bar{X} - \mu}{\frac{\text{standard}}{\sqrt{n}}}$

- confidence interval
  [Lower upper]
  [unknown]

$\Rightarrow \bar{X} \pm$ estimator error
$\Downarrow$
table value $* \dfrac{\text{st dev}}{\sqrt{n}}$

- Lower: $\bar{X}$ - estim
- upper: $\bar{X}$ + estim

- point estimate
  [unknown]

$\Rightarrow \bar{X}$

P-value

more than
- $H_0 \rightarrow$ accept
- $H_1 \rightarrow$ reject

Less than [α]
- $H_0 \rightarrow$ reject
- $H_1 \rightarrow$ accept

P-value

one tail

Right
$*PV = P[Z \geq \text{test stat}]$

Left
$PV = P[Z \leq \text{test stat}]$

two tail
$*PV = 2 \times P[Z \geq \text{test stat}]$

ex $n = 400$ μ = unknown $\sigma^2 = 1600$ $\sigma = \sqrt{1600} = 40$ $\bar{X} = 2485$

- Point est: $\bar{X} = 2485$
- confidence interval = $\bar{X} \pm$ table value $* \frac{\text{St dev}}{\sqrt{n}}$ (given 95%)

$$2485 \pm \underbrace{1.96}_{\text{Given}} + \underbrace{\frac{40}{\sqrt{400}}}_{\text{estimate error}}$$

lower: $2485 - 1.96 * \frac{40}{\sqrt{400}} = 2481$

upper: $2485 + 1.96 * \frac{40}{\sqrt{400}} = 2488$

" $P[2481 < \mu < 2488] = 95\%$ "

Hypo:

$H_0: \mu = 2400$

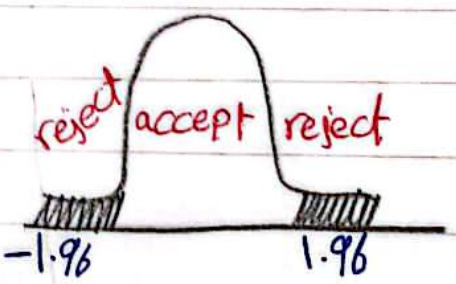
$H_1: \mu \neq 2400$

$\alpha = 5\%$

test stat: $Z = \frac{\bar{X} - \mu}{\frac{\text{St dev}}{\sqrt{n}}} = \frac{2485 - 2400}{\frac{40}{\sqrt{400}}} = 42.5$ جين الـ statistic عشان هيشان كده قولنا reject

table value: two tail = 1.96 (Given)

$H_0 \rightarrow$ reject



- P-value :

$$= 2 \times P[Z > \text{test static}]$$

$$= 2 \times P[Z > 42.5]$$